Project 5 - Foundation Models
CSCI 4521
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100 Sports: Athletics Classification Using Foundation Model Methods

Section 1: Dataset and Problem Definition

The objective of this project is to create a robust system for fine-grained image classification across 100 distinct sports categories. The core challenge lies in the high inter-class similarity of many sports movements and elements; for example, the model must distinguish between highly similar sub-disciplines such as "figure skating men" vs. "figure skating pairs," or "balance beam" vs. "uneven bars." This problem is interesting since it mimics real-world computer vision requirements for tasks such as automated sports broadcasting, player performance analytics, and digital asset management. I chose this problem to explore the performance gap between a specialized foundation model (DINOv2) and the zero-shot capabilities of a general intelligence Large Language Model (LLaVA).

The project utilizes the *100 Sports Image Classification* dataset hosted on Kaggle: https://www.kaggle.com/datasets/gpiosenka/sports-classification

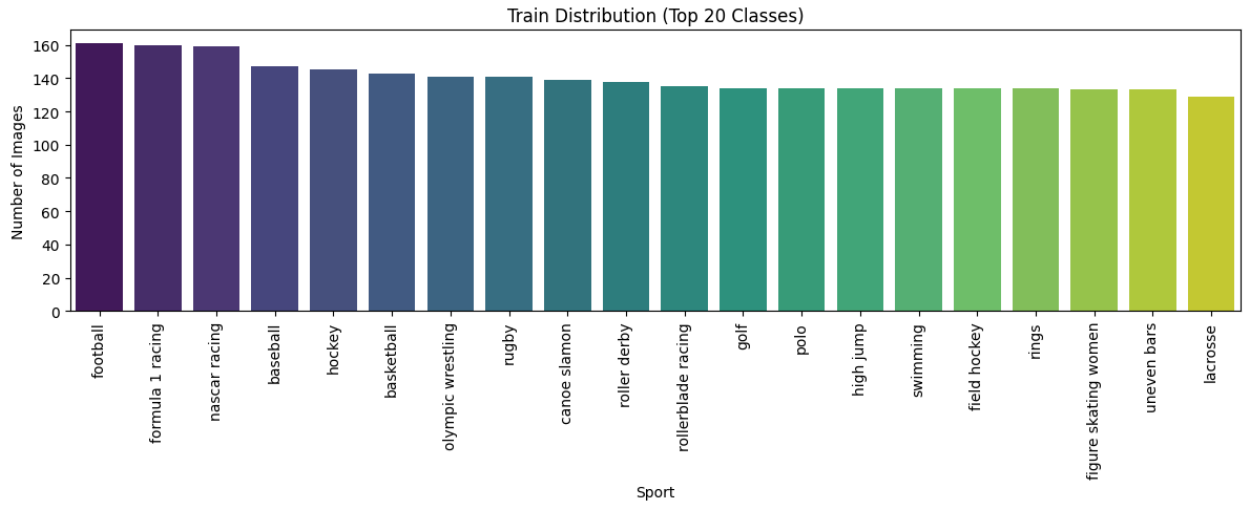


Figure 1. Train Set Class Distribution (Top 20 Classes)



Figure 2. Train Set Class Distribution (Bottom 20)

This dataset is composed of 14,493 high-quality images relatively balanced across all 100 classes as seen in Figures 1 and 2, with most classes containing approximately 100-150 images. Classes with fewer samples appear to have some association with being more obscure, niche sports such as gaga, wheelchair basketball, and sky surfing.

The original dataset folders initially provide testing and validation sets with 500 samples each, with the remaining images comprising the training set. For this project, the data was repartitioned into an 80/10/10 train-validation-test split, which created 11,594 training, 1,449 validation, and 1,450 testing samples, in order to provide a larger testing pool. A notable limitation of this dataset is that most images feature clear, well-lit, and centered action shots. Such clean image data may cause generalization troubles when the model is applied to real-world, low-quality, or blurry sports footage. To address this, several data augmentations were applied to simulate angled camera shots, bad lighting, and motion blur.

Section 2: Embedding Model + Transfer Learning

For the pre-trained embedding model, I utilized DINOv2-Small, which produces embeddings with a dimensionality of 384. With this frozen backbone, I implemented a Multi-Layer Perceptron (MLP) classification head. The MLP architecture takes the 384 embeddings as inputs and consists of two hidden layers (512 units and 128 units respectively), each with ReLU activation and 20% dropout.

The model was trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 256. I implemented an early stopping mechanism with a patience of 6 epochs based on validation loss. As seen in Figure 3 below, the training successfully converged after 14 epochs, reaching a best validation loss of approximately 0.0994.

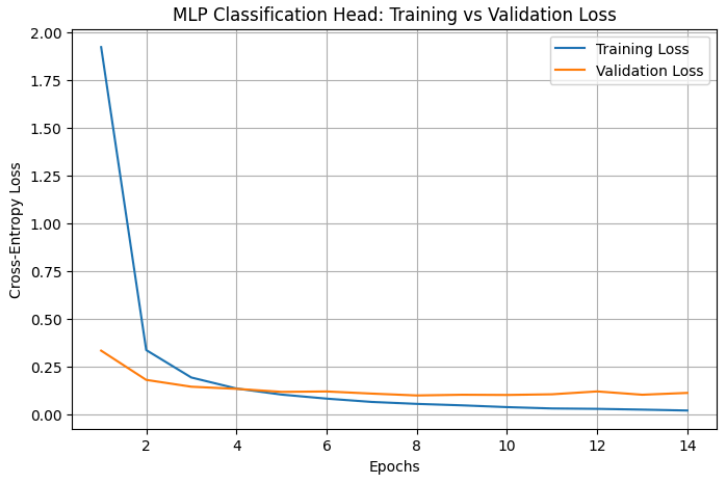


Figure 3. DINOv2 + MLP Training and Validation Loss

Section 3: Large-Language Model

To establish a general intelligence LLM for comparison, I used LLaVA 1.5 (7B), an open-source multimodal Large Language-and-Vision Assistant which comprises approximately 7 billion parameters. The model was run locally within the Colab environment using the HuggingFace transformers library, utilizing 4-bit quantization to fit the model within the VRAM limits of the T4 GPU. I selected LLaVA 1.5 due to its free, open-source accessibility, efficient training, and resource-friendly nature compared to more powerful APIs. Using zero-shot prompting to rely purely on the foundation model's pre-existing knowledge, I submitted the following prompt formatted to match LLaVA's expected schema:

"USER: <image>
You are a professional sports classifier. Look at the provided image and identify the sport being played. You MUST select exactly one sport from the following valid list: [{valid_classes_str}]

1. Output NOTHING except the exact sport name.
2. Be specific! (e.g., choose 'balance beam' or 'parallel bar' instead of just 'gymnastics', choose 'figure skating women' instead of 'figure skating').
3. Do not add any punctuation or extra words.
ASSISTANT:"

In my attempts to tune and improve results, I introduced explicit constraints and repeatedly tried to restrict the prompt. I implemented a string manipulation and Regular Expression (RegEx) pipeline to better extract the final class label, utilizing Python's re module and substring matching, but still extracted many failed "Unknown" classifications. Although tuning appeared to help performance marginally, the model had a persistent tendency to output conversational filler and I was unable to proficiently parse the verbose responses.

Section 4: Evaluation

The models were evaluated using Accuracy, macro-F1 to measure across all 100 classes, and Matthews Correlation Coefficient (MCC) to account for slight class imbalance and high-dimensional label space. Including the improperly extracted class predictions rendered the LLaVA with an accuracy of 0.4750, macro-F1 of 0.3908, and MCC of 0.4917. However, since 145 samples were still properly extracted, any "Unknown" classifications were removed for evaluation.

Metric	DINOv2 + MLP	LLaVA 1.5 (7B)
Accuracy	0.9669	0.6552
Macro-F1	0.9657	0.5055
MCC	0.9666	0.6535

Table 1. Evaluation Results

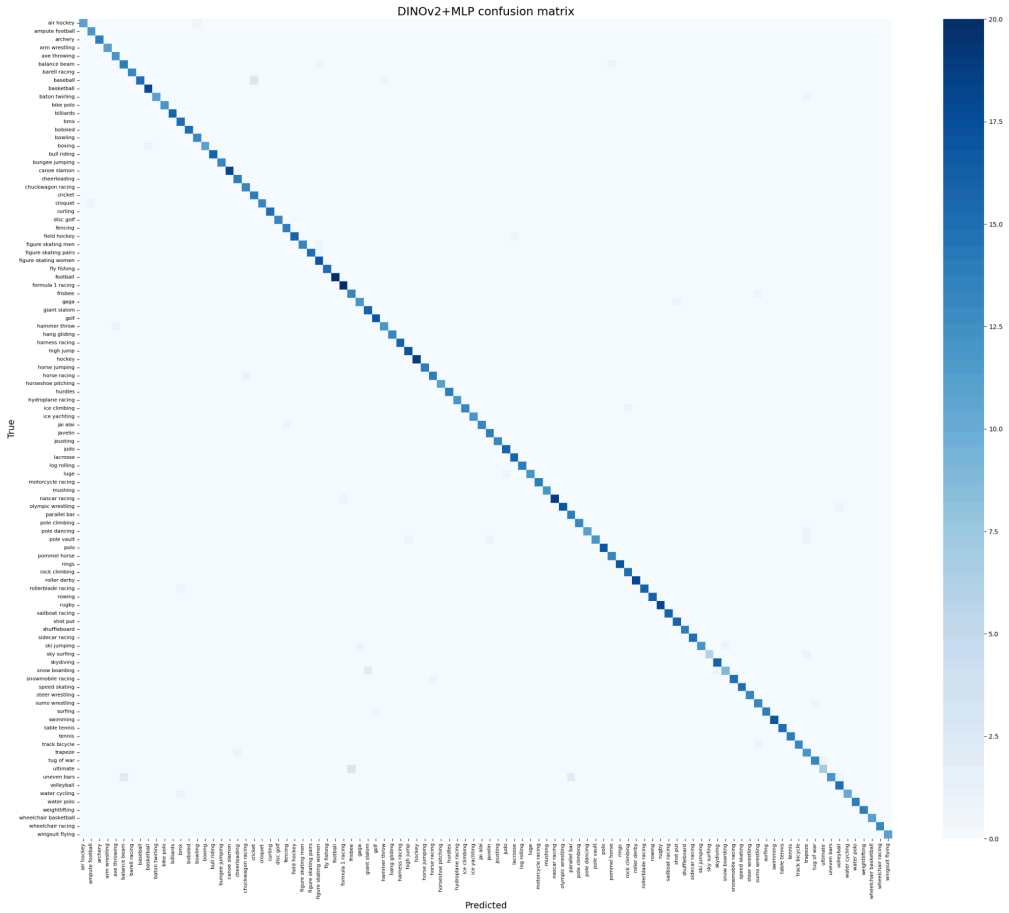


Figure 4. DINOv2 + MLP Confusion Matrix

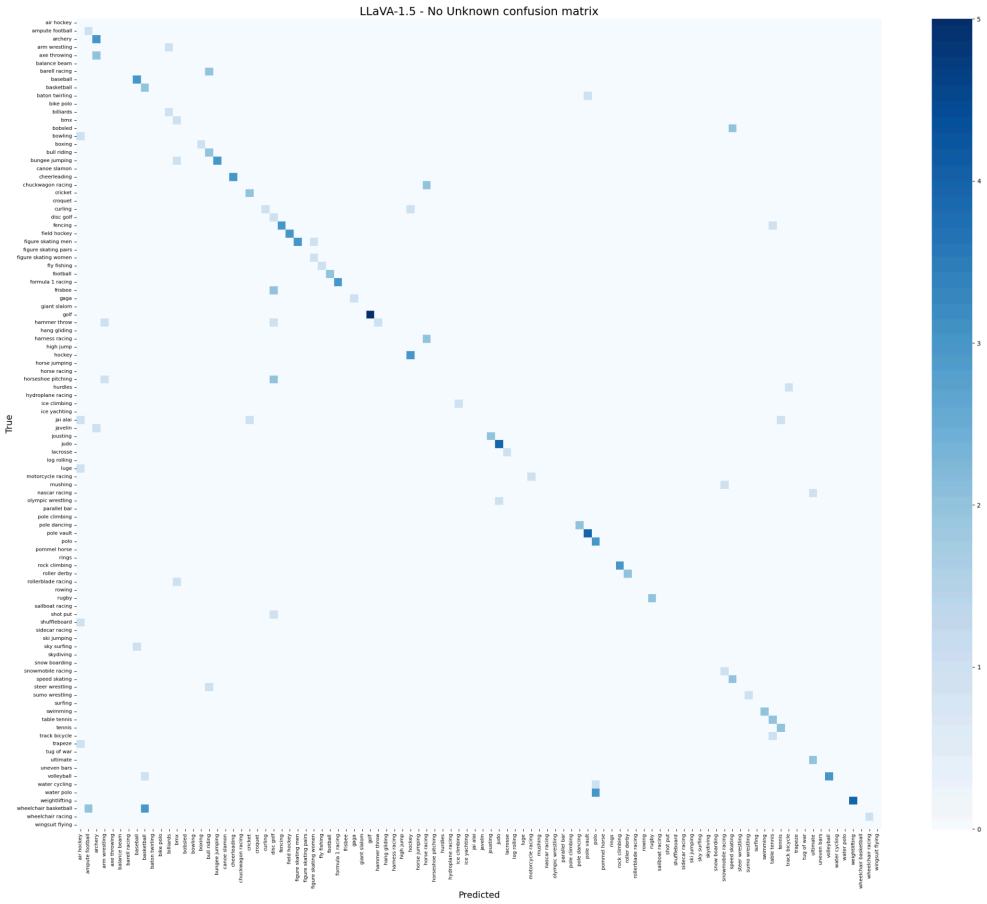


Figure 5. LLaVA 1.5 Confusion Matrix

The DINOv2 pipeline performed exceptionally well in all metrics, even on data augmented by random rotations, color jittering, and Gaussian blurs. The LLaVA model performed significantly worse compared to DINOv2 + MLP, with a lower accuracy by about 31% and lower MCC by about 0.31. However the LLM still performed moderately well, far surpassing a majority-class prediction baseline, which would have achieved slightly better than 0.01 accuracy for this mostly balanced dataset.



Figure 6. DINOv2 + MLP Misclassifications



Figure 7. LLaVA 1.5 Misclassifications

When analyzing failure cases for the DINOv2 + MLP model in Figure 6, we see that misclassifications can be attributed to similar equipment and environments between different sports, For example, “figure skating pairs” and “figure skating women” only differ slightly and “ultimate” vs. “frisbee” both use frisbee discs. Certain movements also obscure equipment, such as the motion of a baseball swing being akin to a cricket swing or the bat resembling a hammer. Some cases also feature more extreme and distanced camera positions that confuse the model. For example, one image displays sumo wrestling with ring ropes from far above and is labeled as tug-of-war. Another image presents a low-angle shot of a snowboarder in midair, which the model confuses for skydiving. This suggests that the DINOv2 + MLP pipeline heavily prioritizes structural, spatial, and silhouette-based features over deep semantic context. If

a snowboarder's spatial orientation mimics a skydiver, or a bat shares the geometric structure of a hammer, the model struggles.

Observing failure cases for the LLaVA model in Figure 7 exhibits similar causes of misclassification, such as "figure skating men" vs. "figure skating women" and "wheelchair basketball" vs. "basketball". It appears to also have more difficulty distinguishing size and scale, especially of equipment, compared to DINOv2. We see that a small baton is mistaken for a large vaulting pole and a large bowling ball is confused for an air hockey paddle. However, some of the examples also demonstrate errors between much more far-removed sports, such as "luge" and "air hockey," which are vastly different in environment and movements. This suggests that LLaVA relies primarily on broad semantic concepts and linguistic priors rather than fine-grained geometric awareness. Its difficulty with scale implies a lack of absolute depth perception. In the cases of "water polo" and "polo" being misclassified twice, despite polo being much different and played on horseback. Due to the similarity of the class label wording, this error is possibly due to imperfect parsing of the LLM response.

For real-world applications and production environments requiring real-time performance (e.g. live sports broadcasts), I would recommend the DINOv2 + MLP implementation due to its superior, balanced performance and immediate class label interpretability. Due to the variety of sports provided by the 100 Sports Image Classification, the model also provides considerations for para athletes as well. An LLM approach, particularly with LLaVA 1.5, is likely better suited for unstructured search or descriptive tagging where a fixed label set is not required.